

CHAPTER 7

Physical Theory of Meteors

7-1. The classical physical theory of meteors. The physical phenomena of meteors are closely interwoven with the properties of the upper atmosphere. Prior to the era of rocket flights, Whipple (1938, 1943) was able to derive amazingly accurate estimates of air densities and temperatures at meteoric heights from his photographic observations of meteor luminosities and decelerations. His calculations were based on what we shall term the classical theory of meteors, which combines conventional aerodynamic ballistics with plausible assumptions about atomic collisions and light-emission processes.

We are now passing through a transitional period and approaching the time when the rocket data (see Appendix) should become sufficiently well established to enable the problem to be worked in reverse. This is particularly important because in recent years we have been forced to the conclusion that most meteoroids in the visible range are porous or crumbly objects, which do not follow the rules for solid mineral bodies. The success of the classical theory appears to have been due in part to the fortuitous selection of very bright meteors in the early camera programs, meteors which were large objects, though not necessarily solid ones, penetrating deeply into the atmosphere. When the same treatment is applied to the majority of visual meteors, a significant lack of agreement may be remarked between predictions and observations.

The discussion in this chapter will be based mainly on the luminous aspects of meteors. One important manifestation of the meteor's passage, the ionization, will receive more detailed attention in Chap. 8. For three reasons we shall devote some time to an outline of the classical theory, despite its limitations. First, it serves quite satisfactorily for some of the brighter meteors at least. Second, the developments in meteor ionization theory (Chap. 8) have stemmed directly from the classical optical theory. The ionization theory seems to have worked quite well in many applications, whether or not it ought to have in the light of current views on the nature of meteoroids. Third, although the "neoclassical" theory of the crumbling meteoroid is being built around

the older hypotheses, it is not yet susceptible of complete delineation in the form of convenient equations—it is in a state of continuous flux, of cut-and-try, and its tenets take the form of empirical tables subject to change.

The classical theory is largely the work of several eminent scientists during the period 1920–1940, including Lindeman and Dobson (1923), C. M. Sparrow, H. B. Maris, and J. Hoppe. Perhaps the greater share of the credit should go to two men: E. J. Öpik, who, in addition to being an indefatigable observer, has displayed a keen insight into the details of the physical processes, and F. L. Whipple, who has supplied accurate observational material and further streamlined the theory. The interested reader may find the historical highlights well covered in several summary papers, notably by Öpik (1937, 1955c, 1955d), Whipple (1938, 1943), Levin (1956), and Herlofson (1948a). Much of the following development of the classical theory is derived from these same papers, though with a distinct leaning toward Whipple's form of presentation, which in turn is based on pioneer work by Hoppe and Öpik.

7-2. The drag and differential mass equations. To begin with, the shape, mass, and density of the meteoroid are all unknown. We may define a dimensionless *shape factor* A such that $A(m/\rho_m)^{2/3}$ is the effective cross-sectional area of the object, where m is the mass and ρ_m the effective density of the body. A simple calculation will show that $A = (9\pi/16)^{1/2} = 1.2$ for a sphere. For a cube $A = 1.0$ to 1.7 , depending on its orientation. Long narrow bodies may have $A < 1$ for the streamlined aspect and $A > 1$ for the broadside-on aspect. In general, irregularly shaped bodies may have a mean A of the same order as that for the sphere, owing to rotation. We shall assume A to be close to unity.

Air particles colliding with the meteoroid will decrease its forward momentum. In time dt the mass of air intercepted by the body will be $dm_a = A(m/\rho_m)^{2/3}\rho_a V dt$, where ρ_a is the air density. The air particles in this volume will gain $\Gamma V dm_a/dt = \Gamma A(m/\rho_m)^{2/3}\rho_a V^2$ units of momentum per second. Here Γ is a dimensionless quantity, known as the *drag coefficient*, which depends chiefly on the shape of the meteoroid, though it may be affected to some extent by other factors such as V and H and whether or not the particle is rotating. Values for Γ vary from 0.5 to 1.0. The meteoroid will lose $m dV/dt$ units of momentum per second, which may be equated to the momentum gained by the air particles, to yield the *resistance or drag equation*

$$\frac{dV}{dt} = - \frac{\Gamma A}{m^{1/3} \rho_m^{2/3}} \rho_a V^2 \quad (7-1)$$

The kinetic energy of the meteoroid is converted to heat, light, and ionization by the collisions with air particles, though the fraction which

appears as heat greatly exceeds the other two. Atoms of the meteoroid are vaporized from the surface of the parent body. Various processes for the ablation of atoms may be visualized, which are conditioned by the composition and size of the meteoroid and the height at which the action is taking place. When the mean free path of the air particles is greater than the radius of the meteoroid, the atoms leave the body without serious mutual interference. Lower down in the atmosphere a shielding effect may arise, owing to a layer of vaporized atoms about the meteoroid. Lower still in the atmosphere, and particularly for larger bodies, an air cap may develop ahead of the meteoroid. These considerations are fully developed by Öpik (1958a). Whatever the mechanism may be, we shall assume that the rate of loss of mass is proportional to the kinetic energy transferred to the intercepted air mass. Thus we may write down the *differential mass equation*

$$\frac{dm}{dt} = -\frac{\Lambda A}{2\zeta} \left(\frac{m}{\rho_m}\right)^{\frac{3}{2}} \rho_a V^3 \quad (7-2)$$

where ζ (dimensions L^2, T^{-2}) is the heat of ablation of the meteoroid material (usually taken to lie in the range 0.2 to 1.0×10^{10} ergs/g), and Λ is the dimensionless *heat-transfer coefficient*. The heat-transfer coefficient is a measure of the efficiency of the collision process in converting kinetic energy to heat: values of Λ in the range 0.6 down to 0.1 or less have been proposed.

7-3. The luminosity and ionization equations. The ablated atoms are moving with the velocity of the meteoroid, and hence their kinetic energies will range from a few tens to some hundreds of electron volts. For example, if we adopt an atomic weight of 30 as a mean value for atoms of stony meteoroids, such an atom moving at 60 km/sec will possess a kinetic energy of $\frac{1}{2} \times 30 \times 1.67 \times 10^{-24} \times (60 \times 10^3)^2 = 9 \times 10^{-10}$ erg, which is equivalent to $(9 \times 10^{-10}) / (1.6 \times 10^{-12}) = 560$ ev. This kinetic energy is more than adequate to excite and also to ionize the atoms of the meteor, for which typical excitation or ionization potentials range from about 2 to 15 ev. The excited atoms emit radiation. If we consider the visible light radiated in the range $4,000$ to $7,000$ Å (see Sec. 2-2), we may define a dimensionless *luminous-efficiency factor* τ_I by assuming that the luminous power I radiated in this wavelength band is proportional to the rate of loss of the kinetic energy of the ablated atoms:

$$I = -\frac{1}{2} \tau_I \frac{dm}{dt} V^2 \quad (7-3)$$

Substituting for dm/dt from Eq. (7-2), we have the *luminosity equation*

$$I = \tau_I \frac{\Lambda A}{4\zeta} \left(\frac{m}{\rho_m}\right)^{\frac{3}{2}} \rho_a V^6 \quad (7-4)$$

The factor τ_I is not easy to evaluate. Various investigators have used estimates which have varied by two orders of magnitude from $\tau_I = 2 \times 10^{-2}$ to 2×10^{-4} . Öpik believes τ_I to be the sum of three factors, the first representing the impact efficiency, the second the temperature-radiation efficiency, and the third the black-body radiation efficiency; of these, the first is the only important component for the average visual meteor. Many years ago Öpik evaluated τ_I from laboratory evidence and theoretical considerations based on the laws of atomic interactions. His conclusions can best be expressed in sets of empirical tables, since the calculated τ_I values depend on the meteor velocity, on the substance of the particle, e.g., whether it is iron or stone, and on whether the meteoroid is a solid body or a dustball. In this brief account we cannot attempt to reproduce all the tables and the reasoning behind them: the reader may be referred to the book by Öpik (1958) for full details. A convenient, though rather sweeping, simplification of these data was produced by Whipple, in the form

$$\tau_I = k_0 V \quad (7-5)$$

where $k_0 = 8.5 \times 10^{-6}$ sec/km. This relation is applicable to the brighter photographic meteors only, and even in these cases its validity is now regarded skeptically. For fainter meteors as seen by the eye, τ_I appears to vary more slowly than the first power of V —perhaps even as a negative power of V for faint dustballs.

The power going into the production of ionization, like the luminous power, is also assumed to be proportional to the kinetic power loss of the ablated atoms. If q is the number of electrons (or ions) produced per unit path length and η (dimensions M, L^2, T^{-2}) is the mean ionization potential per atom involved, then the energy of the ionization created per second is

$$qV\eta = -\frac{1}{2} \tau_q \frac{dm}{dt} V^2 \quad (7-6)$$

where τ_q is the dimensionless *ionization-efficiency factor*. Again substituting for dm/dt from Eq. (7-2), we obtain the *ionization equation* for the number of electrons created per unit path length,

$$q = \tau_q \frac{\Lambda A}{4\zeta\eta} \left(\frac{m}{\rho_m}\right)^{\frac{3}{2}} \rho_a V^4 \quad (7-7)$$

From the combined radio and visual data on Geminid and Perseid meteors, Millman and McKinley (1956) found that the ratio τ_q/τ_I (which is the same as the ratio $qV\eta/I$) varies from about unity for $V = 35$ km/sec to nearly three for $V = 60$ km/sec; hence τ_q/τ_I varies approximately as V^2 (see Sec. 8-13). Whipple (1955a) reached the same conclusion from an independent study of radio and photographic rates.

An alternative path to the ionization equation was followed by the Jodrell Bank workers, who defined the dimensionless *probability factor* β as the probability that a single ablated atom, with a mass μ , would produce a free electron on collision with an air particle. A given meteor atom can, of course, produce more than one electron before it is slowed down to thermal velocities through multiple collisions. The mass of the ions produced per second will be proportional to the mass lost by the meteor per second; that is,

$$qV\mu = -\beta \frac{dm}{dt} \quad (7-8)$$

It follows immediately from Eqs. (7-6) and (7-8) that the link between τ_q and β is given by

$$\tau_q = \frac{2\eta}{\mu V^2} \beta \quad (7-9)$$

Hence, from the above observational evidence that $\tau_q/\tau_I \sim V^2$, one may conclude that $\beta/\tau_I \sim V^4$. If τ_I varies as V^n , as was suggested earlier, then β will vary as V^{4+n} . The Ottawa data (Millman and McKinley, 1956) yielded values that ranged from $V^{3.5}$ for fifth-magnitude meteors to $V^{4.5}$ for zero-magnitude meteors. Hawkins (1956c), using similar observational information, found that on the average $\beta/\tau_I \sim V^{4.56}$. On the other hand, Kaiser (1953) and Evans and Hall (1955) have concluded that β is of the order of 0.1 and is practically independent of the velocity. Further investigations will be needed to resolve these apparent anomalies. In the meantime, the bulk of the evidence appears to support the relations $\beta \sim \tau_q V^2 \sim \tau_I V^4$.

An alternative ionization equation defining q in terms of β and other parameters of the meteor may be derived by substituting for dm/dt in Eq. (7-8) from Eq. (7-2) as follows:

$$q = \beta \frac{\Lambda A}{2\xi\mu} \left(\frac{m}{\rho_m}\right)^{3/2} \rho_a V^2 \quad (7-10)$$

Note that Eq. (7-4) is a power equation, whereas Eqs. (7-7) and (7-10) have the dimensions of inverse length despite their superficial similarity to Eq. (7-4).

7-4. The integrated mass equation. The four basic equations developed in the previous two sections, together with their ancillary equations, may be manipulated to yield a number of interesting and useful results. The deceptively simple elegance of some of these derivations should not cause the reader to forget the tenuous nature of the underlying hypotheses.

The directly observable variables in these equations are V , t , dV/dt , and I . The electron line density q may be included in this list, though it

might better be regarded as an indirectly observed variable deducible from directly observed quantities through the application of a theory which is in process of being established (see Chap. 8). The mass of the meteoroid at any instant is an unknown variable, and its density is a parameter which is also unknown. Several of the remaining parameters, e.g., A , Γ , ξ , may be tentatively assumed to be known to better than an order of magnitude for the average meteor if not for a specific individual, and furthermore they may be considered constant to a first approximation over most of the meteor's flight. Other parameters, such as τ_I , τ_q , and β , have not been well determined, and they may be far from constant. Rather than attempt to measure each unknown separately—there are simply not enough independent equations to do this—we shall group them into certain combinations or ratios which can be evaluated from the observational data. One example of this technique has been mentioned in the previous section, where it was pointed out that the ratios τ_q/τ_I or β/τ_I can be found, though the absolute values remain indefinite.

We may integrate Eq. (7-3) from a given instant t on the light path to the time t_e when the meteor disappears at the end of its flight, to obtain the mass m_t at time t :

$$m_t = \frac{2}{\tau_I V^2} \int_t^{t_e} I dt \quad (7-11)$$

For this purpose the change in velocity right up to the end of the meteor may be neglected, and V^2 may be brought out from under the integral sign. Note here that the product $m_t \tau_I$ can be measured in terms of observable quantities but that neither one alone is determinable. Equation (7-1) is now cubed and the mass therein replaced by the value given in Eq. (7-11) to yield

$$(\Gamma A)^3 \frac{\tau_I}{\rho_m^2} = - \frac{2}{\rho_a^3 V^3} \left(\frac{dV}{dt}\right)^3 \int_t^{t_e} I dt \quad (7-12)$$

The parameters on the left may thus be evaluated as an ensemble in terms of known and observable quantities on the right.

Another observable combination of parameters may be found by dividing Eq. (7-2) by Eq. (7-1) and then substituting for m from Eq. (7-11) and for dm/dt from Eq. (7-3). This gives

$$\frac{2\Gamma\xi}{\Lambda} = - \frac{V}{I} \frac{dV}{dt} \int_t^{t_e} I dt \quad (7-13)$$

We have taken it for granted that the properties of the upper atmosphere are known, e.g., ρ_a in Eq. (7-12), even though a perusal of the tables in the Appendix will show that the values may be uncertain by a factor of two in some cases. We shall now take a further step here and assume

that the atmosphere at meteoric heights may be described as isothermal and homogeneous. This is not true over a great range of height, but it may be taken as approximately correct for small height differences. The air density in an isothermal atmosphere is given by $\rho_a = \rho_0 \exp(-H/H)$, where ρ_0 is sea-level density and H is the scale height at the true height H (see discussion in the Appendix). For a meteor coming from a radiant at a zenith distance Z , the incremental path length is $ds = dH/\cos Z$. Also, we have $V dt = ds$. Equation (7-2), when integrated from $s = -\infty$ on the path, where the mass has the no-atmosphere value m_∞ , to the point s , becomes

$$m^{1/2} = m_\infty^{1/2} - \frac{\Lambda A}{6\zeta \rho_m^{3/2}} \int_{-\infty}^s V^2 \rho_a ds \quad (7-14)$$

Carrying out the integration on the assumption of constant velocity we have

$$m^{1/2} = m_\infty^{1/2} - \frac{\Lambda A V^2 H}{6\zeta \rho_m^{3/2} \cos Z} \rho_a \quad (7-15)$$

as one form of the *integrated mass equation*. Another formulation, from which the atmospheric density and the meteoroid density have both been eliminated, may be derived by again dividing Eq. (7-2) by Eq. (7-1) and integrating directly.

$$\log_e \frac{m_\infty}{m} = \frac{\Lambda}{4\Gamma\zeta} (V_\infty^2 - V^2) \quad (7-16)$$

It will be noted that the same combination of parameters appears in both Eqs. (7-16) and 7-13).

7-5. Maximum luminosity and ionization. As the meteoroid descends into denser air, the production of both luminosity and ionization increases to a maximum and then falls off as the size of the meteoroid dwindles. The condition for maximum light may be found by differentiating Eq. (7-4) logarithmically and equating to zero. We may operate similarly on Eq. (7-7) to determine the condition for maximum ionization. In the latter case, for example, the steps go as follows:

$$\frac{1}{q} \frac{dq}{dt} = \frac{1}{\tau_q} \frac{d\tau_q}{dt} + \frac{2}{3m} \frac{dm}{dt} + \frac{1}{\rho_a} \frac{d\rho_a}{dt} + \frac{4}{V} \frac{dV}{dt} = 0$$

The mass term in this expression becomes $(-\Lambda A \rho_a V^2)/(3\zeta m^{1/2} \rho_m^{3/2})$ on substituting for dm/dt from Eq. (7-2). The air-density term becomes $V \cos Z/H$ with the adoption of a homogeneous isothermal atmosphere. By assuming $\tau_q \sim V^n$, the remaining two terms may be combined into $(4+n)(dV/dt)/V$, which, except for very slow meteors, may be neglected in comparison with either the meteor-mass term or the air-density term.

These latter terms may now be equated to yield

$$\rho_{a(\max)} = \frac{3\zeta \cos Z}{\Lambda H A V^2} m_{(\max)}^{1/2} \rho_m^{3/2} \quad (7-17)$$

where the subscript (max) refers to the values at the point of maximum line density, $q_{\max}$. The identical relation will be obtained for the air density at the point of maximum luminosity, starting with Eq. (7-4). A similar equation is derivable immediately for $p_{a(\max)}$, the atmospheric pressure at maximum light or ionization, since in our assumed isothermal homogeneous atmosphere we have $p_a = gH\rho_a$ (see Appendix), where g is the acceleration due to gravity:

$$p_{a(\max)} = \frac{3g\zeta \cos Z}{\Lambda A V^2} m_{(\max)}^{1/2} \rho_m^{3/2} \quad (7-18)$$

Substitution of the expression for $\rho_{a(\max)}$ from Eq. (7-17) in Eq. (7-15) gives

$$m_{(\max)} = \frac{8}{27} m_\infty \quad (7-19)$$

that is, at the point of maximum light or ionization, the idealized meteoroid has shrunk to $\frac{8}{27}$ of its original mass, or its radius has become $\frac{2}{3}$ of the initial radius.

If we put the value of $\rho_{a(\max)}$ given by Eq. (7-17) into Eq. (7-1) we obtain for the deceleration at maximum light or ionization

$$\left(\frac{dV}{dt}\right)_{(\max)} = -\frac{3\Gamma\zeta \cos Z}{\Lambda H} \quad (7-20)$$

Note that this is independent of meteor size or velocity. The deceleration at maximum light should not be confused with the maximum deceleration, of course, which occurs near the terminal point.

We shall now deduce the connection between the meteor velocity and the height of maximum light or ionization. For meteors of a given density class and an average path inclination to the vertical, we may write the proportionality $\rho_{a(\max)} \sim m_{(\max)}^{1/2}/V^2$, from Eq. (7-17). From Eq. (7-4) we have $I_{\max} \sim \tau_I m_{(\max)}^{3/2} \rho_{a(\max)} V^6$. Eliminating $m_{(\max)}$ gives

$$\rho_{a(\max)} \sim \frac{I_{\max}^{1/2}}{\tau_I^{1/2} V^3} \quad (7-21)$$

Similarly, from Eq. (7-7),

$$\rho_{a(\max)} \sim \frac{q_{\max}^{1/2}}{\tau_q^{1/2} V^{3/2}} \quad (7-22)$$

If we adopt $\tau_q \sim \tau_I V^2 \sim k_0 V^3$, these equations become

$$\rho_{a(\max)} \sim \frac{I_{\max}^{1/2}}{V^{19/4}} \quad (7-23)$$

and

$$\rho_{a(\max)} \sim \frac{q_{\max}^{1/2}}{V^{11/4}} \quad (7-24)$$

If desired, $\rho_{a(\max)}$ may be written for $\rho_{a(\max)}$ in the above equations.

Let us now make use of the relation $\log_{10} \rho_a = 1.3 - (3/40)H$ [Eq. (A-5)], where H is in kilometers and ρ_a in kilograms per cubic meter. On taking logs and substituting for $\log_{10} \rho_a$, we have

$$H_{(\max)} = \text{constant} + 44 \log_{10} V + 1.8 M_{\max} \quad (7-25)$$

where we have inserted $\log_{10} I_{\max} \sim 0.4 M_{\max}$ from Eq. (2-2). Two-figure accuracy seems adequate for the coefficients in view of the various uncertainties. Similarly, Eq. (7-24) leads to

$$H_{(\max)} = \text{constant} + 49 \log_{10} V - 4.4 \log_{10} q_{\max} \quad (7-26)$$

The differential variation of height with V alone has been calculated from Eq. (7-25) for a datum level at $V = 40$ km/sec. This is plotted on Fig. 5-9d as the dashed curve. The agreement is reasonably good between the theoretical curve and the empirical curve based on the observations. A loss of brilliance of 5 magnitudes results in a theoretical height increase of slightly less than 9 km, from Eq. (7-25). This is of the same order as that observed photographically (see Fig. 5-11). The same height increase should theoretically be obtained for a decrease in q by a factor 10^{-2} , though the observations do not appear to bear this out. The radio data in Fig. 5-11 show little height variation with magnitude because these data are based on the heights of the enduring echoes, not on the true heights of maximum ionization. The observed height of the enduring echo will tend to lie between 93 km and the height of maximum ionization.

If we had kept track of all the parameters involved in the above discussion, we should have arrived at numerical values for the constants in Eqs. (7-25) and (7-26). There would have been many uncertain quantities to be evaluated in the process, and it might be better to determine empirical values for these constants directly from the observational data. For example, from Fig. 5-11 (which has already been adjusted to $V = 40$ km/sec) we note that the maximum light curve goes through $H = 93$ km, $M_{pm} = 0$. Since $44 \log_{10} 40 = 70$, it follows that constant = 23 is a reasonable value for Eq. (7-25). Obviously the theoretical straight line between H and M can only approximate to the observed relation.

Similarly, we may use the same fiducial point for the radio echoes, since

at $H = 93$ km, where $M_{pm} = 0$, the point of maximum ionization coincides with the height of maximum echo endurance. Because of the color-index shift in Fig. 5-11, we have $M_v = 1.8 \simeq M_r$, when $M_{pm} = 0$. From Eq. (2-3), $q = 2 \times 10^{15}$ when $M_r = 1.8$. Equation (7-26) then becomes

$$H_{(\max)} = 82 + 49 \log_{10} V - 4.4 \log_{10} q_{\max} \quad (7-27)$$

The variation of $H_{(\max)}$ with velocity alone has been superimposed on Fig. 5-10 as the dashed curve for comparison with the radio observations.

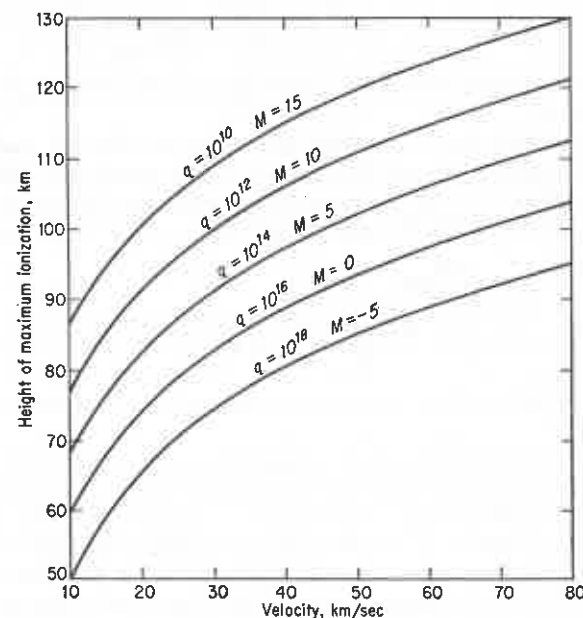


FIG. 7-1. Height of point of maximum ionization as a function of velocity, plotted for selected values of q in electrons per meter, or for the equivalent radio magnitudes.

The theoretical dependence of $H_{(\max)}$ on V has been plotted in Fig. 7-1 for several values of q (or M). These curves will be of interest in assessing the height limitation on the detectability of underdense echoes (Sec. 8-6).

7-6. Variation of mass, light, and ionization. In Eq. (7-15) it will be noted that the coefficient of ρ_a is equal to $m_{\infty}^{1/2}/[3\rho_{a(\max)}]$, from Eqs. (7-17) and (7-19). Hence, the mass at any point is given by

$$m = m_{\infty} \left(1 - \frac{\rho_a}{3\rho_{a(\max)}} \right)^3 \quad (7-28)$$

in terms of the air density at that point, the air density at maximum light or ionization, and the original mass.

We may write down from Eq. (7-4), together with either Eq. (7-7) or (7-10), the following relative relations, making use also of Eq. (7-19):

$$\frac{I}{I_{\max}} = \frac{q}{q_{\max}} = \frac{\rho_a m^{\frac{1}{2}}}{\rho_{a(\max)} m_{(\max)}^{\frac{1}{2}}} = \frac{9}{4} \frac{\rho_a}{\rho_{a(\max)}} \frac{m^{\frac{1}{2}}}{m_{(\max)}^{\frac{1}{2}}}$$

It then follows immediately from Eq. (7-28) that

$$\frac{I}{I_{\max}} = \frac{q}{q_{\max}} = \frac{9}{4} \frac{\rho_a}{\rho_{a(\max)}} \left(1 - \frac{\rho_a}{3\rho_{a(\max)}}\right)^2 \quad (7-29)$$

The density symbols are directly replaceable by the corresponding pressure symbols.

Equation (7-29) is the classical expression for the rise and fall of meteoric luminosity or ionization—often called simply the *light curve*. It can be restated in terms of meteor magnitudes and heights by making use of Eq. (2-2) and the relation $\log_e \rho_a = \log_e \rho_0 - H/H$. Thus,

$$\begin{aligned} \log_e I - \log_e I_{\max} &= \frac{2.3}{2.5} [M_{\max} - M] \\ &= \log_e \frac{9}{4} + \frac{H_{(\max)} - H}{H} + 2 \log_e \left[1 - \frac{e^{(H_{(\max)} - H)/H}}{3}\right] \end{aligned} \quad (7-30)$$

The magnitude variation may also be expressed as a function of time, if desired.

These idealized light curves show that the relative variation in light or ionization along the path should be the same for all meteors; that is, there should be no dependence on meteor mass or density, on meteor velocity, or on path inclination. The absolute height of maximum light of a given meteor will depend on M , V , and Z , of course, and in these respects the theory given in the previous sections of this chapter is in fair agreement with the observations presented in Chap. 5. The ionized trail length (the length of path over which the ionization exceeds an arbitrary threshold level) is also dependent on M , V , and Z . The brighter and faster meteors create longer trails, as do meteors arriving from radiants near the horizon. The Stanford group have concluded that, on the average, an ionized trail length of 25 km is typical of a sixth-magnitude meteor (Manning, Villard, and Peterson, 1953; Eshleman, 1957).

Figure 7-2 shows three typical examples of observed light curves (solid lines) upon each of which theoretical light curves (dashed lines) have been superimposed. The agreement is good in Fig. 7-2a and quite reasonable in Fig. 7-2b despite the multitude of small-intensity outbursts or flares. In Fig. 7-2c the coincidence has been remarkable up to the moment when

a large flare prematurely ended the meteor's career. Very bright meteors such as these usually adhere fairly closely to the simple theoretical curve except for anomalous flare behavior. However, when the observations of fainter meteors are checked against theory, a significantly different pattern develops.

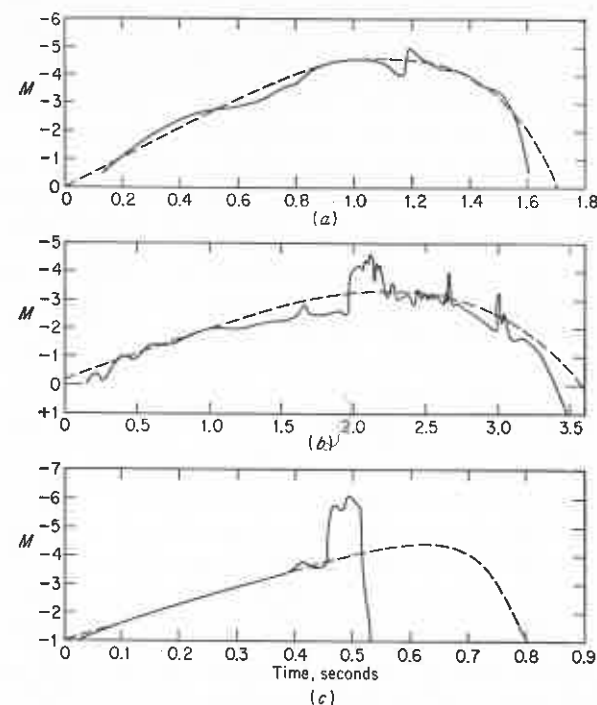


FIG. 7-2. Solid lines, observed light curves of three bright meteors; dashed lines, theoretical light curves. Velocities: (a) 36 km/sec; (b) 25 km/sec; (c) 69 km/sec. (Courtesy of L. G. Jacchia.)

7-7. The fragmentation hypothesis. Hawkins and Southworth (1958) made a random selection of some 360 meteors from the Harvard collection of Super-Schmidt photographs. Many of these were second- and third-magnitude meteors. The magnitude of each was measured at three points on the trail: at the beginning and end where the trail was just detectable, and at the point of maximum light. The data are plotted in Fig. 7-3. For each meteor two points are shown, which represent the differences between peak and beginning magnitudes and between peak and terminal magnitudes; each is plotted against the corresponding true

height differential referred to height at maximum light. The theoretical light curve (Eq. 7-30) is also shown. Practically all of the observations lie inside the theoretical curve. This is strong evidence that the lengths of the fainter photographic meteor trails are considerably shorter than the classical theory predicts.

Jacchia (1955), after examining the light curves of faint photographic meteors, commented that the early part of the light curve frequently displays appreciable irregularities. This is in contrast to the relative smoothness of the initial rise for a very bright meteor. Many faint

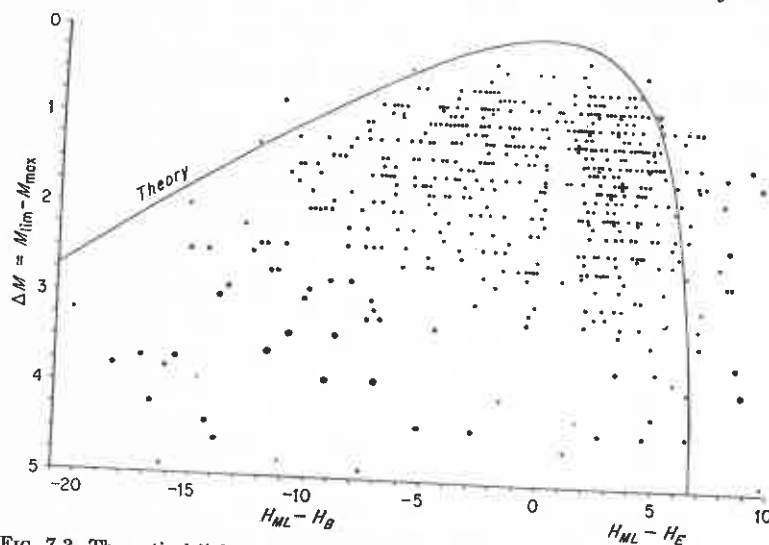


FIG. 7-3. Theoretical light curve compared with the beginning and end points of a random selection of Super-Schmidt meteors. H_{ML} = height at maximum light; H_B = height at beginning of trail; H_E = height at end of trail. (Hawkins and Southworth, 1958.)

meteors show a rapid rise of light followed by a steady decline—a reversal of the classical trend. The shutter breaks in the photographed trail of a faint meteor are not clear-cut. Sometimes the exposed segments become increasingly elongated and fuzzy until, near the end of the trail, they tend to fuse together (McCrosky, 1958). Furthermore, the observed deceleration of a faint meteor often increases much too fast to be explicable by the classical equations of motion for a solid meteoroid.

These observations can be explained if we assume that the average meteoroid has a fragile structure which breaks up easily into a cluster of small fragments on impact with the upper atmosphere. The small

component particles will spread over an appreciable cross-sectional area larger than that of the parent body. There will be a longitudinal spread as well, particularly near the end of the trail, which would account for the shutter-break blurring. If the ultimate particles are true solid bodies, then presumably the classical equations could be applied to each one, but two difficulties arise. First, neither the numbers nor the range of sizes of the component particles are known. Öpik has estimated that the radius of an individual grain may lie in the range 0.001 to 0.05 cm. The number originally packed together in the parent meteoroid may run from 10 to 1,000 for ordinary dustballs and to over a million for the Giacobinid meteors. The second difficulty is that the particles are released in an abnormal environment, at lower heights and in denser air than they would have reached as independent bodies; hence their individual light curves will be appreciably shortened. The presence of many other particles in close proximity also affects the analysis. Practically all dustballs—which is to say almost all meteors—are now thought to break up into grains when the aerodynamic drag $\rho_a V^2$ reaches a value of about 2×10^4 dynes/cm² $\approx$ 0.02 atmosphere. The effective density ρ_m of the parent meteoroid may be of the order of 0.05 to 0.5 g/cm³, which would lead to an increase of nearly two orders of magnitude in the estimated masses of the meteors.

Enough has been said here to underline the new uncertainty that has crept into the physical theory of meteors. At one time, the neat and tidy classical theory was made to serve for all meteors, to a first order of approximation. The advent of more precise optical measurements has shown up its shortcomings, though the iconoclastic process has in many instances merely replaced old anomalies with fresh ones. The effect of the fragmentation hypothesis may be felt even more strongly in the theory of radio echoes from meteors, which has been tied very closely to the classical concept of a single solid body.

The treatment in this chapter has been mainly phenomenological in nature. We have not delved deeply into the basic physics and chemistry of the problem. It would be unwise to attempt to deal superficially with the intricate details of the collision and ablation processes, sputtering, the "jet" effect, shock-wave phenomena, and the various theories of energy transfer. The interested reader may be directed to the books by Öpik (1958a) or Levin (1956), or to the series of papers on these subjects from Harvard (e.g., Cook, 1954; Jacchia, 1955; Whipple, 1955a), but he should be cautioned that an adequate grounding in fundamental physics would be most helpful, if not indeed essential, to a critical comprehension of those treatises.